Grade 10 – Term 1
Choosing a Decision Criterion: C10 Practice

Evaluated Criterion – C10

Evaluates economic and financial decision-making scenarios by selecting and justifying the most appropriate decision criterion according to the information available, risk attitude, and decision context.

1 Your Assignment

Write an **individual PDF article analyzing a real-life economic or financial decision under uncertainty**. Use the concepts developed in C1–C9 to connect the decision context, financial information, and recommendation. **The main purpose of C10 is to justify your choice of decision criterion, not merely to perform calculations.** Select and defend one appropriate criterion; you are not required to calculate results for all four criteria or reproduce C9's staged filtering procedure.

2 Prepare Your Analysis

1. **Choose a specific, realistic situation.** Consider a household purchase with uncertain running costs, a small business's stock decision, or an event's spending plan with uncertain attendance. Identify an actual decision, a time period, and financial consequences that you can investigate. Choose a manageable situation with at least two feasible alternatives and two possible states of nature; a broad topic such as "investing money" is not sufficiently specific.
2. **Describe the context.** State the problem, who makes the decision, and what they want to achieve. Distinguish alternatives (actions the decision-maker controls) from states of nature (uncertain conditions beyond their control). Define mutually exclusive states that cover the relevant possibilities in your model. Explain the decision-maker's risk attitude and constraints, such as budget, acceptable loss, available time, or a minimum financial requirement. Explain any exclusions of infeasible alternatives before applying your criterion.
3. **Build a payoff table.** Put alternatives A_i in rows and states S_j in columns. Each entry $P_{ay}(A_i, S_j)$ must show the financial consequence of that combination. Use consistent currency, units, and time period. Prefer net profit, net gain, or savings so that larger payoffs are better; negative values represent losses. If your sources give costs, explain how you convert them into comparable net payoffs. Identify relevant revenues and costs and show how you obtained the entries. Use realistic, verifiable information where possible, such as published prices, supplier quotations, or historical records. **Distinguish sourced facts from assumptions and estimates.** Cite factual data; label and justify estimates and assumptions, including how you derived them. Do not present an uncertain future payoff as an established fact.
4. **Select and defend your criterion.** Consider Maximax, Maximin, Minimax Regret, and Expected Monetary Value (EMV/Bayes). Explain which best fits your information, objectives, risk attitude, and constraints. Use the brief guide below to support your reasoning, rather than choosing whichever criterion produces your preferred alternative.
    - **Maximax:** highest row maximum; appropriate when pursuing the greatest possible upside is the priority and the downside is acceptable.
    - **Maximin:** highest row minimum; appropriate when protecting the worst-case financial outcome is the priority.
    - **Minimax Regret:** smallest maximum regret; appropriate when limiting missed opportunities is the priority. For each state, regret is its best feasible payoff minus the alternative's payoff.

- **EMV/Bayes:** highest probability-weighted average payoff; appropriate when defensible state probabilities are available and a risk-neutral monetary objective fits the context. An expected value is not a guaranteed outcome.

The first three criteria do not require probabilities. For EMV, provide and justify a probability for every state; probabilities must be nonnegative and sum to 1. **Do not invent probabilities or assume equal probabilities simply because they are unknown.** If probabilities cannot be justified, select a criterion that does not require them. Even when probabilities exist, explain whether an expected monetary result fits the decision-maker's ability to bear losses.

5. **Apply your selected criterion.** Show the relevant calculations for each feasible alternative and identify the recommended alternative, including any ties. For Minimax Regret, include the regret table and each alternative's maximum regret; for EMV, show the probability-weighted sums. Interpret the result in the original currency, units, and context.
6. **Justify the recommendation and its limits.** Explain why your criterion is more suitable than the other three for this situation; a short conceptual comparison is sufficient. Connect your reasoning to the evidence and constraints, not only to a risk-attitude label. Acknowledge what the criterion ignores, uncertainty in your estimates, and whether a plausible change in an important assumption could alter your recommendation.

3 Required Article Structure

Organize your analysis as a connected article using the following headings. Integrate tables and calculations into your explanation; do not submit a collection of disconnected answers.

Title and author Give a descriptive title and your full name.

Introduction Present the real-life situation, decision-maker, decision problem, and objective.

Methodology Describe alternatives, states, financial information, sources, assumptions, estimates, constraints, and your selected decision criterion.

Results and Discussion Include the labeled payoff table, necessary calculations, interpretation, and justification of the criterion relative to the alternatives. Discuss limitations.

Conclusion State your final recommendation and explain how the analysis supports it.

References List the sources used for contextual information and financial data, when applicable. Include enough detail to locate them (author or organization, title, date, and link where available), and cite them where used.

4 Before You Submit

Submit your individual article as a **readable, professionally organized PDF**. Use clear headings, complete paragraphs, consistent notation, and legible tables with currency and units. Check that calculations support the recommendation, facts are distinguishable from estimates, and the reasons for selecting your criterion are explicit. Open the exported PDF to check that equations and tables are complete and readable.